Cross-Domain Interface Catalogue

Synthetic AUTOSAR-inspired High-Level Design Knowledge Document

Important: This document is fictional, independently created for academic evaluation, and contains no confidential OEM data or reproduced AUTOSAR standard text.

Field	Value
Document ID	SYN-CAT-IF-001
Version	1.0
Status	Approved synthetic reference
Classification	Synthetic academic test data
Intended use	RAG, OCR, extraction, comparison, and evaluation
Authority	Not valid for production design or vehicle release

1. Interface naming rules

Interface names use the prefix If_. Provided ports use Pp and required ports use Rp. Data-element names use descriptive CamelCase names followed by a unit suffix when a numeric unit is applicable.

2. Cross-domain interface catalogue

Interface	Type	Provider domain	Data elements	Nominal cycle
If_VehicleSpeed	Sender-Receiver	Powertrain	VehicleSpeed_kph, MotionState	10 or 20 ms by revision
If_EngineState	Sender-Receiver	Powertrain	EngineRunning, EngineSpeed_rpm	20 ms
If_DoorState	Sender-Receiver	Body Control	DoorAjar_FL, AllDoorsClosed	50 ms
If_LockStatus	Sender-Receiver	Body Control	LockState	100 ms
If_EventStatus	Sender-Receiver	Diagnostics	EventId, EventStatus	On change
If_FreezeFrame	Client-Server	Diagnostics	SnapshotId, SnapshotPayload	On request

3. Relationship constraints

Every required port shall reference a declared interface. Each sender-receiver interface shall have exactly one logical provider in this synthetic architecture. A provided port may have zero or more consumers, but unused interfaces shall be reported for review.

4. Source precedence

Priority	Source type	Rule
1	Approved revision-specific HLD	Overrides older revisions for the same project
2	Approved interface catalogue	Used when the HLD does not redefine the interface
3	Component catalogue	Provides identity and ownership
4	Review candidate	May be queried but must be visibly marked as unapproved

5. Citation requirement

Answers shall cite the source file, page number, and section. If evidence is absent, the system shall return insufficient evidence rather than infer a relationship.

6. Controlled duplicate

If_VehicleSpeed appears in both revision-specific HLDs. This is valid across versions and must not be treated as a same-version duplicate when the document version filter is applied.